IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Lean Six Sigma & Continuous Improvement

Reading notes for the morning session. Module 3 — Methods, Tools and Operational Efficiency, Session 4.

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Prof. Dr. Thiago Lima

International Certificate in Project Management and Organizational Performance

No preparation is required. These notes carry the definitions, figures and sources, so the time in the room can be spent on judgement rather than on terminology.

Contents

The session builds one argument in twelve movements: value is defined from the customer backwards, waste is what the customer would not pay for, and the efficiency that matters is the unit's journey rather than the resource's occupancy.

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> **Bring three things.** One recurring process that takes longer than it should; two rough figures — projects in flight, and projects closed in the last twelve months; and one authorised project you know well enough to defend, which is the raw material for the group activity in Part Nine.

Most organisations that adopt lean copy the tools. Few reproduce the results. These notes build the argument for why — and the argument runs on three equations rather than on a list of techniques.

They open with a puzzle. **The answer is deliberately not in this document.** You will be asked to write your hypothesis down in the first ten minutes of the session, and we will test it against the framework at the end of the morning — so if you are reading ahead, stop at the end of Part One and let the morning do the rest.

> **WHAT THE SESSION WILL ASK OF YOU**
>
> Nothing to prepare, but three things to have in mind — all of them about your own organisation, all answerable from what you already know.
>
> **One recurring process** you own or depend on, that takes longer than it should. Contract approval, project initiation, monthly close, incident resolution.
>
> **Two rough figures:** how many projects are in flight in your unit, and how many closed in the last twelve months. Approximations are fine.
>
> **One authorised project** you know well enough to defend. It closes the session as the group activity in Part Nine. A project you have private doubts about is better material than one you are proud of.

A note on framing. The examples below run through factories more often than through project offices, because that is where the ideas were developed and tested. They were not developed for factories. Each section ends with a **bridge to project management**, because the mechanisms transfer exactly — queues are queues, variability is variability — and because this programme is about delivery rather than manufacturing.

ONE

The puzzle

Fremont, California

1982. General Motors closes its Fremont assembly plant. Absenteeism runs near 20%. The workforce is represented by what GM managers describe as the most combative union local in the system. Quality is the worst in General Motors.

1984. The plant reopens as NUMMI, a joint venture with Toyota. Substantially the same workforce is rehired, including the union leadership. Same building. Same equipment. Within roughly two years, quality and productivity approach Toyota's Japanese levels and lead the entire GM system.

Same assets. Same people. **What changed — and where would that change appear on a balance sheet?**
And a second question, harder: GM had access to NUMMI from the inside for two decades. Why was it never replicated across the rest of the network?

Hold your answer. Everything that follows is a way of making it precise.

TWO

Value, seen from the customer backwards

Taiichi Ohno described the whole of the Toyota Production System in one sentence: *"All we are doing is looking at the timeline — from the moment the customer gives us an order to the point when we collect the cash — and we are reducing that timeline by removing the non-value-added wastes."*

That sentence does two things at once. It defines the unit of analysis as **elapsed time from order to cash**, not cost per unit. And it makes the customer, not the accounting system, the judge of what counts as work.

Three kinds of activity

Type	Test	Typical share
Value-adding	Transforms the product or service in a way the customer would knowingly pay for	Under 5%
Necessary non-value-adding	The customer would not pay for it, but law, technology or contract requires it	10–30%
Waste (*muda*)	Neither	The remainder

In value streams that have never been deliberately improved, **value-adding time is a low single-digit percentage of elapsed time.** Everything else is queue. This is the single most consequential number in operations, and almost no organisation measures it.

What the customer is actually choosing

"What the customer would pay for" is not a marketing abstraction. It decomposes into operational variables.

The **competitive priorities** — cost, quality, delivery speed, flexibility — are the operational determinants of willingness to pay. Terry Hill's distinction then sorts them by function. An **order qualifier** is a threshold: its presence attracts nobody, its absence eliminates you from consideration. An **order winner** is what makes a customer choose you once you are in the set. Nobody selects a restaurant because its bathroom is excellent; plenty never return because it was dirty.

The managerial consequence is uncomfortable. Effort poured into qualifiers above the threshold consumes capacity and produces no preference. To be excellent at an order winner, an organisation has to accept being merely adequate elsewhere — and a firm that is mid-table on every attribute has made no choice at all.

David Garvin's eight dimensions of quality sharpen this further, because quality is not one variable. **Conformance** lowers cost; **reliability, durability and serviceability** raise willingness to pay. The same quality effort therefore pays twice, on two different edges of value.

IN PROJECT TERMS

THE IDEA A qualifier buys nothing above the threshold; only a winner creates preference. Spending beyond the threshold on a qualifier is waste with a strategic label.

IN A PROJECT Most portfolio scoring models add strategic fit, NPV, risk and resource availability on one scale — which quietly assumes every criterion is a winner. Compliance and regulatory projects are qualifiers: they buy no advantage and not doing them ends the business. Put thirty initiatives through such a model and the eighteen compliance items, easy to justify and hard to argue against, come out ahead of the twelve that would actually move a customer.

ON MONDAY Split the list *before* scoring. Ask of every item: if we do not do this, do we lose a deal or do we lose the licence? Fund the second kind to the threshold and no further; let only the first kind compete for what is left.

The value stick

Brandenburger and Stuart formalised where value comes from:

$$\text{Value created} = \text{WTP} - \text{SOC}$$

willingness to pay, minus supplier opportunity cost

Between those two bounds sit three claims: customer surplus (WTP − price), firm margin (price − cost), and supplier surplus (cost − SOC). Negotiating over price or over supplier terms moves value *between* parties. Only two moves **create** it: push WTP up, or push SOC down.

Lean is unusual because it does both at once. Removing waste lowers cost; the same work raises reliability and speed, which raise what the customer will pay. That is the whole strategic claim of the method, and the rest of these notes explain the mechanism.

From value created to advantage

Creating value and having an advantage are different things, and the value stick alone cannot tell them apart. A firm that raises WTP and lowers cost in a market where every competitor is doing the same has created value and captured none of the difference. Advantage is a *comparison*, so it has to be written as one:

$$VC = (WTP_A - Cost_A) - (WTP_B - Cost_B)$$

the value firm A creates, minus the value created by its relevant rival B

Competitive advantage is the **difference between the value two firms create** — not the value one of them creates in isolation. Three consequences follow, and each of them changes how an improvement programme should be argued.

- **The benchmark is a rival, not last year.** An improvement that cuts your own cost by eight per cent while the reference competitor cuts theirs by twelve has made the firm better and its position worse. Internal time series cannot detect this; only the comparison can.
- **Both terms can be attacked.** VC rises when A improves and equally when B's advantage is made harder to sustain. Most of the literature in these notes is about the first term. The second is why capability that resists imitation matters more than capability that does not.
- **Operational effectiveness erodes by construction.** If the improvement is visible and copyable, B eventually makes the same move, the two parentheses converge, and VC tends to zero. The firm is left creating more value than before and capturing no more of it than its rival. This is Porter's objection, stated as arithmetic rather than as an opinion, and the afternoon session returns to it as the central risk of benchmarking.

The practical use of the formula is in the third reading. It gives a single test to put to any improvement project: *when the competitor does this too, what is left?* If the answer is nothing, the project is a qualifier and should be funded to the threshold and no further. If something is left, name it — that residue is the advantage.

Where the residue can live

Naming it is easier with the identity that organises the rest of the day. Van Mieghem and Allon describe an operation through four linked elements — value, capabilities, assets and processes — and the relationship between them answers the question VC leaves open.

$$V = C \times (A + P)$$

Lima (2026), Ch. 1 — after Van Mieghem & Allon (2015)

A is what the organisation owns: plants, machines, systems. **P** is how work flows across them. **C** is its repeated ability to improve P. The form of the expression is the claim: assets and processes are *additive and purchasable*, capability is a *coefficient*.

Read against VC, that settles where the difference between two firms can survive. Anything in the bracket is available to a competitor at roughly the same price, so improvements that live there move both parentheses and the difference returns towards zero. Only the coefficient resists, because it is the one term nobody can invoice you for. **This is the whole argument of the day, stated early so that everything after it can be read as evidence for or against it** — and Part Eight returns to the identity once there is enough material to test it.

The five principles, and where each one is treated

Womack and Jones reduced the method to five principles in sequence. They are worth stating as a set once, because the sequence is itself the argument: every principle after the first is meaningless until the first has been answered.

	Principle	What it asks	Where
1	**Value**	What is the customer actually paying for? Defined from the customer backwards, never from the process forwards.	This part
2	**Value stream**	Which activities deliver that value, which merely enable it, and which do neither? The map is the instrument.	Part Three
3	**Flow**	Can the work move without stopping? Stoppage, not slowness, is what makes lead time long.	Part Four
4	**Pull**	Is work released by downstream demand, or pushed by upstream capacity and a plan?	Part Five
5	**Perfection**	Does the cycle repeat, with the standard rewritten each time? This is the principle that decides whether the first four survive.	Parts Seven and Eleven

THREE

The anatomy of waste

Ohno catalogued **seven** wastes, and that is the canonical list; modern practice adds an eighth, unused talent, which is not Ohno's and is the one most relevant to knowledge work. The list matters less than its structure: **one of them is the root, and the others are symptoms.**

Overproduction is the root waste

Producing more than is needed, earlier than it is needed, generates the other six. It creates inventory, which needs transport and motion, which hides defects, which forces waiting when the defect is finally found. Attack overproduction and the others shrink; attack the others individually and overproduction regenerates them.

Waste	In a factory	In project and knowledge work
Overproduction	Making more than demand	Reports nobody reads, features nobody uses, gold-plating
Waiting	Idle machines and people	Approval queues, decision latency, blocked dependencies
Transport	Moving materials	Handoffs between departments and systems
Over-processing	Machining beyond specification	Sign-off chains, redundant status meetings, precision beyond need
Inventory	Stockpiles and work in process (WIP)	Half-finished work packages, open tickets, projects started but unstaffed
Motion	Unnecessary movement of people	Searching for information, tool switching, duplicated data entry
Defects	Scrap and rework	Rework loops, failed deliverables, change requests from unclear requirements
Unused talent	Ignored operator ideas	Experts doing administration; front-line insight never reaching decisions

Making it visible — the value stream map

The list above is a vocabulary. A value stream map is the instrument that locates the items in a specific process, and it is the second Lean principle made operational. It is drawn by walking the process, not by assembling it from what people say in a meeting — which is why the first draft almost always contradicts the official procedure.

A map carries three things at once, and the third is the one that does the work.

Layer	What it records	What it reveals
The work	Each step, in sequence, with who performs it	Handoffs — which are transport, and where the defects enter
The information	What triggers each step: a signal, a schedule, a queue, an approval	Whether the stream is pulled or pushed, and where decisions wait on data
The timeline	For each step, the time actually worked and the time elapsed before the next step starts	The ratio between the two, which is flow efficiency, and is the number the map exists to produce

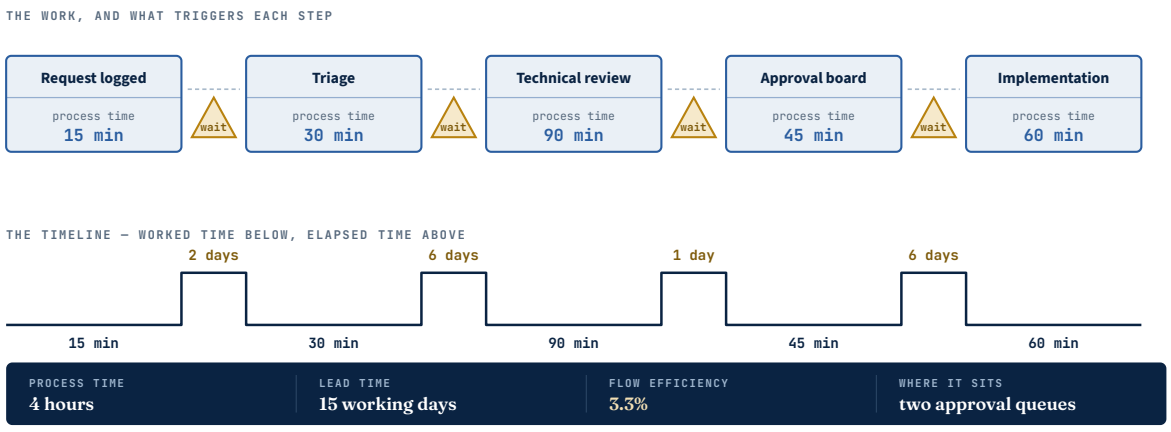


Figure 1. A value stream map of an administrative process — a change request from arrival to implementation. The boxes carry the worked time, the triangles the waiting. The staircase at the bottom is the whole point of the exercise: four hours of work spread across fifteen working days, with almost all of the elapsed time sitting in two approval queues.

Two figures come out of the timeline. **Process time** is the total time anyone is actually working on the unit. **Lead time** is the elapsed time from the first step to the last. In administrative and project processes the first is typically between two and ten per cent of the second, and the gap is not idleness — it is queue. Everything in the waste table lives in that gap.

A useful map adds one more figure per step: the proportion of items arriving **complete and accurate**, needing no clarification or rework. A chain of steps each at 90% complete-and-accurate does not deliver 90%, and the arithmetic of that is in Part Four.

> **CURRENT STATE, THEN FUTURE STATE**
>
> The map of today is diagnosis. The second map — the future state — is the improvement plan, and it is what makes the exercise more than documentation. The discipline is that every difference between the two maps must be attributable to a named change with an owner. A future state nobody can trace back to specific interventions is a wish drawn in boxes.

IN PROJECT TERMS

THE IDEA Map the stream, not the deliverable. The elapsed time lives in the handoffs around the work, not in the work.

IN A PROJECT The streams worth mapping are the administrative ones: change request to approved change, invoice to payment, defect reported to defect closed. They repeat, they can be timed, and they are where lead time accumulates. The characteristic finding is the one in Figure 1 — four hours of work across fifteen working days, with the days sitting in two approval queues that exist because of an authority limit nobody has revisited since it was set.

ON MONDAY Pick one of the three. Walk it with the people who perform each step, not from the procedure. Write two numbers on one line: hours worked, days elapsed. The ratio is the conversation.

Muda is the symptom; mura and muri are the disease

Three Japanese terms, and executives usually learn only the first.

Mura — unevenness

Irregular flow of work. Unevenness creates queues, and queues create the safety buffers that look like prudent management and are in fact inventory.

Muri — overburden

People and equipment pushed beyond reasonable limits. Overburden creates defects, breakdowns and absence — which then create more unevenness.

This is why improvement programmes plateau after year one. The first wave removes visible muda and produces real gains. Then the gains stop, because the unevenness and overburden that generate muda were never touched, and they regenerate it.

IN PROJECT TERMS

THE IDEA Overburden (*muri*) and unevenness (*mura*) are not wastes themselves; they are what manufactures the wastes. Remove the symptoms and they come back.

IN A PROJECT Chronic over-allocation is *muri* with a resource-management label. A specialist at 120% across four projects produces the entire symptom set at once: defects, because attention is divided; waiting, because each project queues behind the other three; unused talent, because the scarce expert spends the day reconciling status. The quarter-end delivery peak that every organisation knows and none levels is *mura* — treated as a fact of the calendar, it is usually a consequence of how work was released.

ON MONDAY List everyone allocated above 100%, and count projects per person. Two is a working limit; four is a queue with a name. Then look at the release calendar for the peak and ask what was decided three months earlier that produced it.

FOUR

The efficiency paradox

Lean is routinely introduced as "doing more with less". Taken literally, that is the definition of the thing lean exists to correct.

Resource efficiency

Keep every person and every machine busy. It optimises what cost accounting measures, which is why organisations default to it.

It generates queues, rework and expediting — and it destroys performance globally while every local unit looks efficient.

Flow efficiency

Optimise the unit's journey through the system. It measures the order, not the resource.

It usually improves both once queues collapse — and almost nobody reports it.

Flow efficiency is simply the share of elapsed time in which the unit is actually being worked on — process time over lead time, the ratio the value stream map in Part Three was built to produce. The figure is usually a shock the first time it is computed, and the shock is the point.

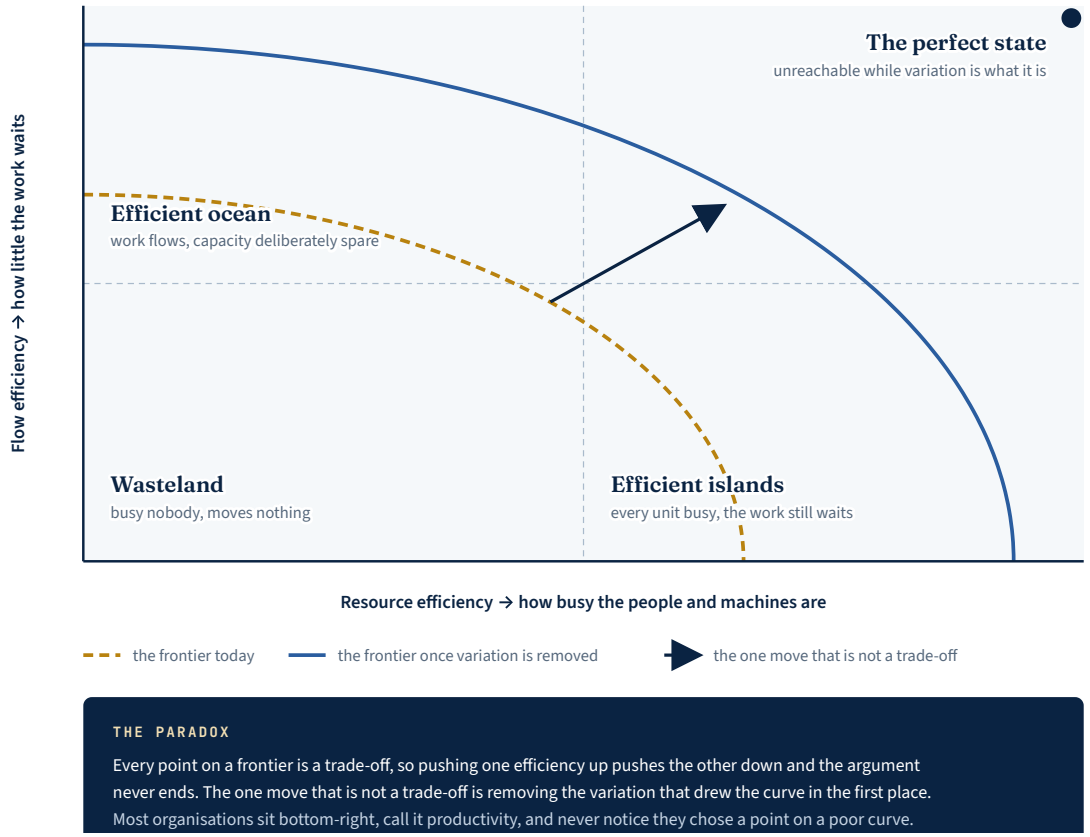


Figure 2. The efficiency matrix, after Modig and Åhlström (2012). Resource efficiency and flow efficiency are not two names for the same thing, and the curve is the reason they trade off: with variation at its current level, the combinations beyond the frontier cannot be bought with effort. Removing variation is what moves the frontier towards the corner where both are high.

The physics of lead time

The relationship between utilisation and queue time is not a management opinion. It is a property of any system with variability, and it is steep:

- As utilisation rises toward 100%, **queue time rises towards infinity**. The curve is flat until roughly 80% and then climbs violently.
- Therefore the instinct to keep every resource busy is **the direct mechanical cause of the inventory it was meant to avoid.**
- Raising utilisation does not raise the system's output. It raises the queue in front of the system.

> **THE SENTENCE WORTH MEMORISING**
>
> **Variability must be absorbed by inventory, by capacity, or by time.** Every organisation chooses one — usually without deciding to. Only one of the three is invisible in the P&L, which is why it is the one most organisations choose by default.

The other half of the method — variation

Everything above is about variability without naming its source, and this is where the Six Sigma half of Lean Six Sigma enters. Lean attacks the time the work spends not being worked on. Six Sigma attacks the **dispersion** of the process — and dispersion is precisely what makes lead time unpredictable. They are not alternative programmes. They are two attacks on the same quantity, which is why the integration is more than a brand.

Two kinds of variation, and only one is a problem to be solved

Shewhart's distinction governs everything that follows. **Common-cause** variation is inherent in the process as designed: it is the ordinary scatter a stable process produces, and no amount of investigating individual cases will reduce it, because every case has a different story and none of the stories is the cause. **Special-cause** variation comes from something that was not there before, and it can be found and removed.

The managerial consequence is uncomfortable and expensive to ignore. **Reacting to common-cause variation as though it were special cause makes the process worse**, because each reaction is an unplanned adjustment to a stable system. The monthly meeting that demands an explanation for every variance against target, favourable or adverse, is this error institutionalised — and it is the single most common misuse of a performance dashboard.



Figure 3. A control chart of the same process over twenty-four measurements. The limits are calculated from the process itself, not from the target. Two signals are present and everything else is noise — and the costly managerial error is not missing the two signals, it is explaining the noise.

Capability — the promise is the spread, not the average

A process has a voice, which is its natural spread, and the customer has a requirement, which is a tolerance. Capability compares the two.

$$C_p = (USL - LSL) / 6\sigma$$

the width the customer allows, over the width the process produces

C_p assumes the process is centred on the target. C_{pk} removes that assumption and is therefore the one that matters, because a process can be admirably narrow and reliably in the wrong place. A conventional threshold is $C_{pk} \geq 1.33$, and the operational reading of it is simple: **the average tells you almost nothing about what the customer will experience; the spread, relative to the tolerance, tells you everything.**

This transfers directly to services and to projects, where the tolerance is usually a promise rather than an engineering specification. A response committed at 48 hours with an average of 30 and a spread that reaches 90 is a process that will break its promise regularly while reporting comfortable averages every month.

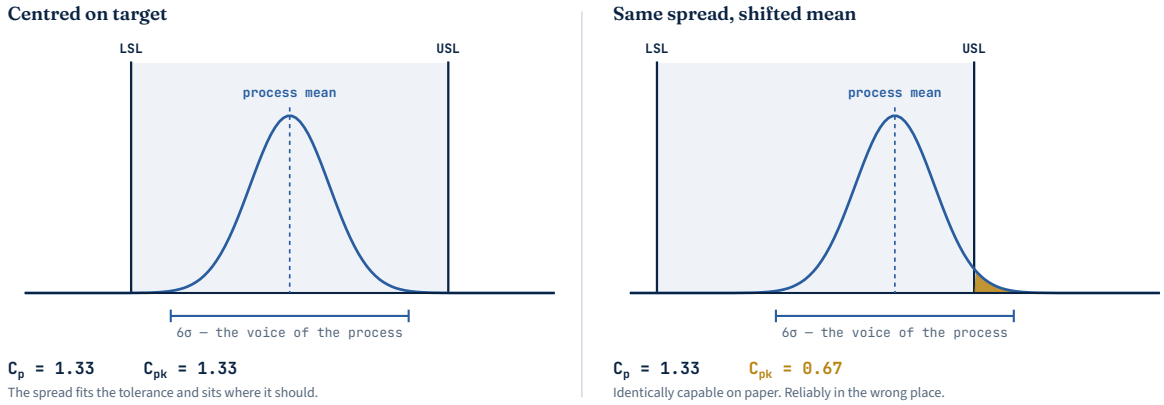


Figure 4. Two processes with identical spread and opposite fates. On the left the mean sits on target and the tolerance is comfortable. On the right nothing about the process changed except where it is centred — C_p is unmoved, C_{pk} halves, and the shaded tail is the share of output the customer will reject. This is why the average is a poor promise and the spread, read against the tolerance, is the real one.

What a sigma level actually says

Sigma level	Defects per million	Right first time	What it means in practice
3σ	66,807	93.3%	Typical of a process nobody has studied. Feels acceptable in isolation.
4σ	6,210	99.38%	A well-run process with active controls.
5σ	233	99.977%	Requires design changes, not vigilance.

6σ	3.4	99.99966%	The benchmark the method is named after.

These figures follow the convention of allowing a 1.5σ long-term drift in the process mean, which is why 6σ corresponds to 3.4 defects per million rather than to the two-per-billion a static normal distribution would give. It is a convention, not a law, and it is worth knowing that it is one.

THE ARITHMETIC THAT MAKES THE CASE TO AN EXECUTIVE

Quality is almost never argued step by step; it is argued at the step and experienced across the chain. A twenty-step process in which **every step is 99% right** delivers a correct outcome **81.8%** of the time. At 99.38% per step — a 4σ process — it delivers 88.3%. At 99.9% per step it delivers 98.0%.

So the familiar objection, *our quality is already good*, is usually true at the step and false at the outcome. It also explains why removing steps is a quality intervention and not only a cost one: every handoff you delete is a term removed from that product.

WHY THE TWO HALVES NEED EACH OTHER

Lean without Six Sigma removes steps and queues and then cannot hold the gain, because the variation that generated the queues is still there and rebuilds them as buffers. Six Sigma without Lean produces beautifully capable steps inside a process whose elapsed time is still ninety per cent waiting — a stable process delivering slowly and predictably.

The connection back to Part Two is direct. Reducing variation lowers cost through less rework, which moves the SOC edge; and it raises reliability and conformance, which moves the WTP edge. It is the same double movement the value stick predicted, now with a measurable instrument attached.

IN PROJECT TERMS

THE IDEA Variability must be absorbed somewhere — in inventory, in capacity, or in time. An organisation that has not chosen has chosen time, because time is the only buffer invisible in the P&L.

IN A PROJECT The three buffers have exact equivalents: inventory is work in progress and half-finished deliverables; capacity is overtime, contractors and reserve; time is schedule contingency. Goldratt's critical chain is the argument that these should be sized deliberately and placed where they protect the whole project — not hidden inside every task estimate, where they are consumed silently and protect nothing. A plan that allocates people at 100% has decided, without saying so, that every illness, late approval and changed requirement will be absorbed by the schedule. The resulting delay is the planning assumption working exactly as specified.

ON MONDAY Open the resource plan and find the allocation assumption. If it is 100%, the schedule is the buffer. Then find where contingency lives: if it is in every task, nobody will ever see it spent.

FIVE

The Toyota Production System, as a system

The TPS is conventionally drawn as a house, and the drawing is an argument: a roof needs pillars, pillars need a foundation, and removing any element collapses the rest.

Goals — highest quality · lowest cost · shortest lead time · best safety · high morale

Just-In-Time	People and teamwork	Jidoka
Takt time · continuous flow · pull and WIP limits · SMED	Kaizen · *genchi genbutsu* · structured problem solving. The centre of the house, and the multiplier.	Autonomation · stop-and-fix · andon · poka-yoke

Stability — *heijunka* (levelling) · standardised work · 4M stability

The terms in the house, in one line each. *Takt time* — the pace of customer demand, the beat the whole line is set to. *SMED* — single-minute exchange of die: changeovers made fast enough that small batches become economical. *Genchi genbutsu* — go and see for yourself, at the place where the work happens. *Autonomation* — a machine that stops itself when something goes wrong. *Andon* — the signal, usually a light or a cord, by which anyone halts the line on a problem. *Poka-yoke* — a device that makes the error physically impossible. *Heijunka* — levelling volume and mix so the line sees a steady load. *4M* — the four sources of variation to stabilise first: man, machine, material, method.

The sequence is physics, not preference. Pull cannot function on an unstable, unlevelled base: it either starves or floods. This is why organisations that install kanban boards — the cards or columns that signal when the next unit may be started — onto chaotic processes get chaos with better visualisation.

The left pillar: Just-In-Time

Three mechanisms, and one widespread misconception to clear first.

> PULL IS NOT JIT, AND NEITHER IS KANBAN
>
> **A pull system is a system that explicitly limits work in process.** That is the definition. A push system releases work according to an external schedule. Kanban is merely one mechanism for limiting WIP; CONWIP is another. Treating the cards as the essence of the system is the most common conceptual error in the boardroom (Hopp & Spearman, 2004).

- **Continuous flow** — remove the interruptions, batching and breakdowns that convert value-adding time into elapsed time.
- **Pull** — produce nothing until a downstream signal indicates demand. The signal is what stops overproduction.
- **SMED** — batch sizes exist because changeovers are expensive. Kill the changeover and the economic batch size collapses towards one.

Underneath all three sits *heijunka*: levelling the schedule so that the pull signal is stable enough to be worth following.

The right pillar: Jidoka

Quality built in at the source, through three mechanisms of increasing cost and decreasing frequency:

- **Poka-yoke** — design the process so the mistake physically cannot be made.
- **Andon** — signal the abnormality the instant it occurs, not at the end of the shift.
- **Stop-and-fix** — halt the line so that the defect never travels downstream.

Stopping the line looks like the most expensive thing a factory can do, and that reading misses the trade. **A problem that stops the line gets solved; a problem that is worked around does not.** The cost of stopping is visible and immediate; the cost of working around is invisible and permanent.

> IN PROJECT TERMS
>
> THE IDEA Most of the Toyota house already exists in a project organisation under other names. One element does not, and it is the one that matters most.
>
> IN A PROJECT *Heijunka* is resource levelling. Standardised work is the methodology, templates and checklists the PMO maintains. *Poka-yoke* is the mandatory field that will not let the form be submitted incomplete. The element with no comfortable equivalent is the andon cord. Monthly status reporting is its opposite: it aggregates upward on a cycle, and a red status costs the reporter more than it costs the organisation — a system designed to signal abnormalities late and quietly.
>
> ON MONDAY Measure one lag: from the moment a problem is first known to someone on the team to the moment it reaches someone who can act. Then ask what happened, the last time, to the person who reported red early. The answer is your andon policy, whatever the governance document says.

Why the system beats the tools

This is the best-evidenced claim in the field, and it is the reason the house is drawn as a house rather than as a menu.

Source	Finding
Milgrom & Roberts (1990)	Modern manufacturing is a system of *complementarities*: the return to any one practice rises with the adoption of the others. Partial adoption yields near-zero or negative returns.
MacDuffie (1995)	Bundles of mutually reinforcing practices explain performance — not individual practices.
Shah & Ward (2007)	Built the field's measurement instruments for lean exclusively on bundle logic.

This is the strongest available explanation for the failure rate of tool-led lean programmes. It also explains something counter-intuitive: a partial implementation can leave an organisation *worse* off than no implementation, because a practice adopted without its complements imposes its costs without delivering its returns.

SIX

Lean in the language of capital

Operational improvement that cannot be stated in financial terms does not survive the budget cycle. Fortunately, the translation is an identity rather than an argument.

Little's Law

$$\text{WIP} = \text{Throughput} \times \text{Lead Time}$$

Little (1961)

Hold throughput constant and the identity says something immediate: **cutting lead time cuts work in process proportionally.** And work in process is not an operational abstraction — it is inventory, and inventory is invested capital.

`lead time ↓` → `WIP ↓` → `inventory ↓` → `invested capital ↓` → **`ROIC ↑`**

Flow improvement is a balance-sheet intervention that merely happens to take place on a shop floor.

IN PROJECT TERMS

THE IDEA Little's Law is an identity, not a model: average time in the system equals the number in the system divided by the rate at which they leave.

IN A PROJECT Take projects in flight as work in progress and projects completed per year as throughput, and the identity returns the average time a project spends in the organisation:

`average project lead time = projects in flight ÷ projects completed per year`

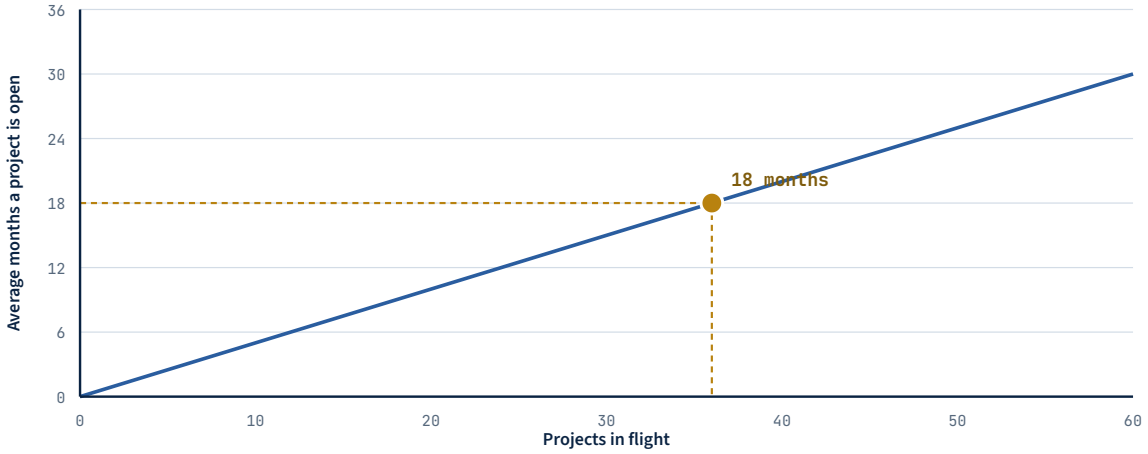
Twenty projects in flight, ten finished a year: a two-year average lead time, whatever the schedules say. Halve the number in flight and the lead time halves, with the same people and the same budget — and throughput usually *rises*, because the context switching that consumed the difference disappears. This is the counter-intuitive law every steering committee eventually rediscovers at cost: **starting less finishes more.**

ON MONDAY Two counts: everything authorised and not closed, and everything closed in the last twelve months. Divide. Put the result beside the lead time your organisation believes it has. The gap is the agenda for the next portfolio review.

Little's Law, as a portfolio instrument

Shown here at its default setting. The version on the course page has sliders, and the chart and the sentence below move with them.

lead time = 36 in flight ÷ 24 finished / year = 1.5 years = **18** months



Start less — in flight 36 → 22

11 months

same people, same budget

Finish more — per year 24 → 29

14.9 months

20% more capacity, bought or hired

To match the first lever by capacity alone

40 / year

a 67% increase in throughput

With **36** projects in flight and **24** finished a year, the average project is open for **18 months**. Cutting the number in flight by 40% brings that to **11**; adding 20% capacity brings it only to **14.9**. The first lever is a decision. The second is a budget.

Figure 5. Little's Law applied to a portfolio: the average time a project stays open is the number in flight divided by the number finished per year. Throughput is set by capacity and is hard to move quickly; the number in flight is a decision somebody takes every week. The law holds on averages, for a system that is stable and where what starts eventually finishes.

READING THE CHART HONESTLY

The relationship is an identity, so there is nothing to forecast and nothing to dispute. What it does require is a stable system in which what starts eventually finishes: a portfolio that quietly abandons a third of what it begins is not described by this line, and the first useful thing such an organisation can do is count the abandonments.

WE DO THIS CALCULATION LIVE IN THE SESSION

A. Projects currently in flight in your organisation or unit. Count everything authorised and not yet closed, including the ones nobody has touched this month. Those count — they are WIP.
B. Projects completed and closed in the last twelve months.

A ÷ B is your real average project lead time. Approximate figures are enough. The interesting number is not the result — it is the gap between that result and the lead time your organisation believes it has.

Where lean shows up in return on invested capital

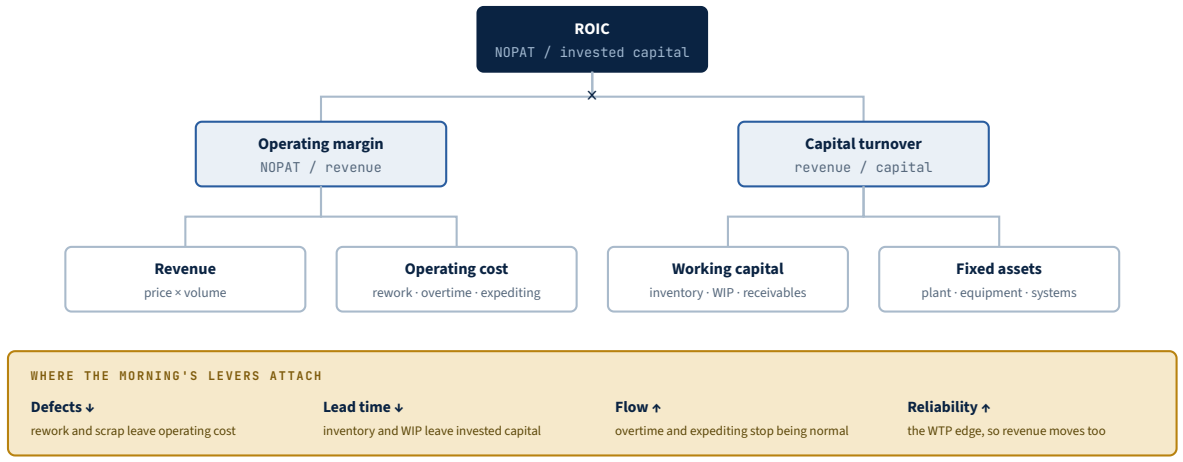


Figure 6. Return on invested capital, decomposed. The identity has two branches and the morning's work lands on both: removing defects and expediting lifts the left branch, and removing lead time lifts the right one by taking capital out of the denominator. A programme argued only as cost saving is claiming half of what it does.

ROIC — return on invested capital — is NOPAT, net operating profit after tax, divided by the capital invested to earn it. It splits into two independent levers:

Margin — the numerator

NOPAT ÷ revenue. Lean moves it through scrap, rework, yield and equipment effectiveness.

This is where every standard lean business case is written.

Capital turnover — the denominator

Revenue ÷ invested capital. Lean moves it through inventory, footprint, capacity buffers and the cash cycle.

This is where nobody in the operations meeting is looking.

A worked scenario

Lead time is cut from 20 days to 8. Defect reduction adds 20 to NOPAT. Revenue is unchanged.

Measure	Before	After
Inventory	300	120
Invested capital	800	620
Operating margin	8%	10%
ROIC	10%	16.1%
WACC — the cost of that capital	10%	10%
Economic profit	0	38

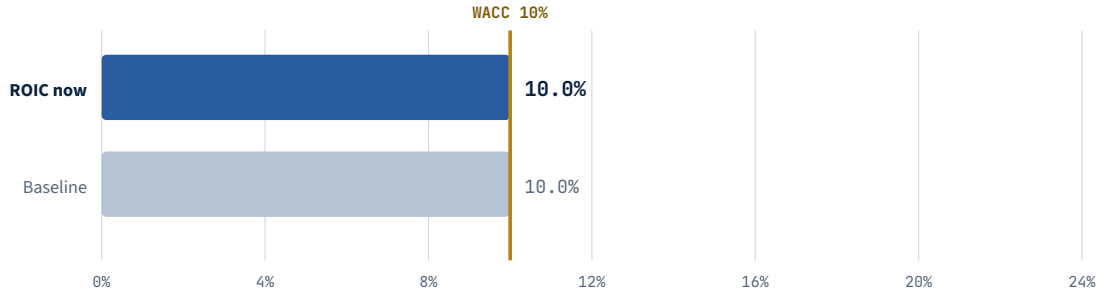
Margin improved by 25%. Capital turnover improved by 29%. **The denominator did more work than the numerator** — and the denominator is the half of the equation that operational business cases almost never mention. A firm that was exactly earning its cost of capital is now creating economic profit, without a single additional unit of revenue.

What an operational improvement does to return on capital

Shown here at its default setting. The version on the course page has sliders, and the chart and the sentence below move with them.

capital = 500 (fixed) + 15 × 20 (inventory) = **800**

ROIC = 1,000 × 8% (NOPAT = 80) ÷ 800 (capital) = **10.0%**



INVENTORY	INVESTED CAPITAL	NOPAT	ROIC	ECONOMIC PROFIT
300	800	80	10.0%	+0

> Return equals the cost of capital exactly. The accounting profit is real and the economic profit is zero.

Scenario	Lead time	Margin	Capital	ROIC	Economic profit
Baseline	**20 days**	**8%**	**800**	**10.0%**	**+0**
Flow only	8 days	8%	620	**12.9%**	+18
Cost only	20 days	10%	800	**12.5%**	+20
Both	8 days	10%	620	**16.1%**	+38

Figure 7. The same improvement, read as a return on capital. Shortening lead time releases inventory, which leaves the denominator; removing rework and expediting lifts the margin, which raises the numerator. The grey bar is the starting position, where return exactly equals the cost of

capital. The gold line is the only threshold that matters. The figures are stylised; in your own numbers, inventory per day of lead time is roughly annual cost of sales divided by working days.

The two inputs are the morning's two mechanisms, and the figure is worth operating at least once with a deliberately unhelpful setting: raise the margin as far as it goes while leaving lead time at thirty days, and watch how much of the improvement the denominator eats. That is the shape of a cost-reduction programme that never touched flow.

Both pillars move both edges of value

Mechanism	Effect
Jidoka → cost	Defects removed at source; the cost of poor quality collapses. The classic business case.
Jidoka → WTP	Zero-defect delivery lets a B2B customer eliminate incoming inspection and cut safety stock. A willingness-to-pay argument with a hard number attached.
JIT → cost	Less inventory, handling, expediting and obsolescence; a smaller footprint.
JIT → WTP	Customers pay for speed and predictability, because short lead times cut their own forecasting risk.

SEVEN

DMAIC — conducting improvement, analysing problems

Everything so far describes what good looks like. DMAIC is the method for getting there from where you are — a discipline for conducting an improvement and, above all, for analysing a problem before paying to solve it.

The failure mode it exists to prevent

Six Sigma was developed at Motorola in the 1980s and scaled across industry in the following decade. Its statistical ambition — a process so consistent that specification limits sit six standard deviations from the mean, roughly 3.4 defects per million opportunities — is the part everyone remembers. It is not the part that matters most to a senior manager.

What matters is the discipline. An experienced manager recognises a problem in seconds and already knows what to do. That recognition is pattern-matching against previous problems: fast, frequently right, and entirely unverified. When it is wrong, the organisation finds out after it has paid for the solution — and because the supposed cause was never written down, the error teaches nothing.

DMAIC does not exist to distrust experience. It exists so that the experienced manager's hypothesis is written down before it is funded, and can therefore be refuted while refutation is still cheap.

The five phases

Phase	The question it answers	Representative tools
Define	What is the problem, whose is it, what does it cost, and where does the scope end?	Project charter · SIPOC · voice of the customer
Measure	What is today's performance, measured in a way anyone would trust?	Operational definitions · measurement system analysis
Analyze	What is the cause — verified, not assumed?	Five whys · Ishikawa · Pareto · hypothesis testing
Improve	Does the solution work in a pilot, before it is scaled?	Controlled pilot · poka-yoke · design of experiments
Control	Does the gain survive the dissolution of the project team?	Control charts · control plan · standardised work

The third column names the usual instruments of each phase; it does not teach them, and none of them is examined here. What matters is the question in the second column — a phase is complete when its question has an answer somebody would stake a decision on, whichever instrument produced it.

Two observations that are easy to skip. First, Define and Measure look like bureaucracy and are half the work: a badly defined problem has no verifiable solution, and a measure nobody trusts ends every discussion in opinion. Second, **poka-yoke appears in this table and also in the Jidoka pillar above.** Lean and Six Sigma are not rival systems; they share mechanisms and attack different enemies — flow and waste on one side, variation and defects on the other.

Analyze, in practice

This is the phase we work through live in the session, on the queue the value stream sketch in Part Nine reveals.

The five whys is the simplest root-cause technique and the easiest to perform badly. Two rules govern it:

- **A cause attributed to a person is not a process cause.** "Because he did not deliver on time" is a restatement of the symptom. The next question is always: what in the process allows that to happen without anyone noticing?
- **Stop when you reach something somebody controls.** A chain that bottoms out in "because the market is volatile" has gone one step too far; a chain that stops at the second link has not gone far enough.

What the technique reliably demonstrates is that the cause people name first is usually the third or fourth link in the chain — that is, still a symptom. Pareto analysis then answers a different question: of the many verified causes, which few account for most of the effect.

Where initiatives die

In Analyze

The team arrives with its preferred solution and uses the phase to justify it. The diagnostic symptom is simple: the root cause turns out to be whatever someone already wanted to buy.

In Control

The gain exists for as long as the team exists. Once the team disperses, the process regresses within months, because nobody inherited the standard.

The cheapest safeguard against both is a **tollgate** at the end of each phase, chaired by the executive sponsor who owns the business result rather than by the specialist running the project. Whoever conducts the work is not whoever approves it.

IN PROJECT TERMS

THE IDEA DMAIC is a project with the weight in a different place: a conventional plan commits to a solution at authorisation; DMAIC forbids naming one until a cause has been verified.

IN A PROJECT The vocabulary overlaps almost exactly — the charter is a project charter, the tollgates are stage gates, the sponsor is the sponsor. The gate between Analyze and Improve is the one with no equivalent in most governance frameworks, and it is the one that does the work. Control maps onto benefits realisation and handover to operations — the phase every methodology describes and few organisations perform. A project closed with the process owner untrained and the measure unmonitored has reached Improve and stopped.

ON MONDAY At the next approval, ask what the verified cause is. If the answer is a solution — "we need a new system", "we need more people" — the project is at Define, whatever the gate it is passing through says.

Where the method does not apply

DMAIC assumes a repeatable process with a definable defect. Applied to the exploratory front end of a business — research, invention, early-stage product discovery — it suppresses precisely the variation that discovery depends on. The organisation needs two regimes, one for exploring and one for exploiting, and clarity about which process is in which.

EIGHT

The third equation

The identity was stated in Part Two, before there was anything to test it against. Everything since has been that test: the waste, the queues, the variation, the system of complementarities, the method for improving it. This part puts the three terms back together.

$$V = C \times (A + P)$$

Lima (2026), Ch. 1 — after Van Mieghem & Allon (2015)

Term	What it is	Visibility
A — Assets	Plants, machines, systems. What the pillars stand on.	Visible and purchasable
P — Processes	How work flows across the assets. The pillars and the foundation.	Partly visible, imitable with effort
C — Capability	The repeated ability to improve P. The centre of the house.	Barely visible, rarely imitable

The form of the equation is the claim. **Assets and processes are additive and purchasable; capability is a coefficient.** You can buy a plant, license a platform, hire a consultant to redraw a process. You cannot buy the organisation's repeated ability to improve. And if C approaches zero, V approaches zero regardless of what A and P are worth.

IN PROJECT TERMS

THE IDEA Sort what a PMO owns by the three terms, and most of it lands in the bracket.

IN A PROJECT The scheduling tool, the portfolio platform, the dashboards: **A**. The methodology, the templates, the gate model, the certified staff: **P**. Both are purchasable, and both are available to every competitor at roughly the same price. **C** is what is left — whether this organisation reliably gets better at delivering than it was two years ago, and whether that improvement survives the departure of the people who drove it. It is also why two organisations running the same methodology, with the same certifications and the same tooling, deliver at visibly different standards: a gap no maturity assessment fully explains and every experienced manager recognises.

ON MONDAY Three questions for the PMO. What do we deliver better this year than last, and how do we know? Which improvement survived its author leaving? What would remain if the tool were replaced tomorrow? The third answer is C, and it is usually shorter than anyone expects.

The puzzle, answered

Part One asked two questions and told you to hold your answer. The identity answers both.

What changed at Fremont, and where does it appear on a balance sheet? Not A: same building, same equipment. Not only P: the processes changed, but GM photographed and copied those for twenty years without the result. What changed was C — the organisation's repeated ability to find and fix its own problems, with the same people who had been unable to the year before. On the balance sheet it appears nowhere, which is the point. It shows up only in its consequences: in the quality figures, in the inventory that was no longer needed, in a return on capital that no line item explains.

Why was it never replicated across GM? Because what GM could observe was the bracket, and what produced the result was the coefficient. The kanban boards, the andon cords, the standardised work sheets were all visible and all copied. The rules-in-use that made them work — how problems are surfaced, who is allowed to stop the line, how an improvement is run as an experiment — were not written down anywhere, because at Toyota nobody needed to write them down. That is the whole of the afternoon's argument about benchmarking, and it is why the day ends on how C is built rather than on where it can be bought.

The equations, in one place

Equation	What it establishes
Value stick WTP − SOC	Lean pushes both edges outward at once: Jidoka and JIT each lower cost and raise willingness to pay.

Competitive advantage $VC = (WTP_A - C_A) - (WTP_B - C_B)$	The comparison form of the first. Creating value is necessary; creating more of it than the rival is what the firm is paid for. When an improvement is copyable, the two brackets converge and VC returns to zero.
Economic profit ROIC vs. WACC	Lean lifts the numerator and cuts the denominator, moving return above the cost of capital.
VCAP $V = C \times (A + P)$	It does so only for organisations that build the multiplier rather than buying the bracket.

TWO IDEAS TO BRING TO THE SESSION

Porter's objection. Operational effectiveness is imitable, so its gains compete away; strategy-grade advantage must live somewhere harder to copy. Where, exactly — and what does that imply for an improvement programme whose entire output is better processes?

Spear and Bowen's finding. What distinguishes Toyota is not its tools but its *rules-in-use*: tacit rules governing how flow paths are specified and how every improvement is run as a controlled experiment. Tacit rules can be observed and still not transferred. Hold that thought against the puzzle in Part One.

NINE

Group activity — The Value Case

The final block of the session is worked in groups. You will take one real project and appraise it through the three lenses built in these notes: where it moves willingness to pay, where it removes waste, and what it actually builds.

> **THE RAW MATERIAL**
>
> **One authorised project from your own organisation** — in flight or recently approved — that you know well enough to defend. It does not need to be large, successful, or yours. Groups choose one project from among those their members propose, and the choice takes ninety seconds: pick the one the group disagrees about, not the one with the best business case.

Why this appraisal differs from a standard business case

A conventional business case answers one question: does the return exceed the cost? That question is necessary and it is not sufficient, because it is silent on three things that decide whether the return survives.

- **Which edge of value does this move?** A project that only lowers cost is competing on the edge where every rival is also working. A project that raises willingness to pay is harder, rarer, and usually under-argued because the number is harder to produce.
- **Is the cost saving a named waste, or an aspiration?** "Improves efficiency by 15%" is a target. "Eliminates the three re-approvals caused by incomplete submissions" is a waste with a name, and only the second can be verified afterwards.
- **Does the organisation end up more capable, or only better equipped?** A project that delivers assets and processes can be replicated by any competitor willing to spend the same money. A project that leaves the organisation better at improving cannot.

Those three questions are one question in disguise, and VC states it: a project is worth defending to the extent that it widens the gap between *(WTP − cost)* for your firm and the same quantity for the competitor you are actually losing deals to. Step 6 of the canvas is where that is settled.

> **THE INSTRUMENT**
>
> The canvas is a separate page, fillable and exportable to PDF in the same way as the afternoon's appraisal: **The Value Case Canvas** — also em português. Open it now. Groups of three or four complete it on one project in about fifteen minutes, and each group then defends it in three. The page carries its own instructions and a summary of the ideas each field uses; what follows is why it asks what it asks.

Six steps, one argument

Step	What it asks
1 · The project	What it is, who sponsors it, and what it was approved to achieve.
2 · The WTP edge	Which competitive priority it moves; the quality dimension, if any; order winner or qualifier; and whether the customer would notice.
3 · Map the stream	Three to six steps with who performs them; process time; lead time; and where the gap sits.
4 · The cost edge	The waste removed, named from the eight — not a benefit; where it occurs today.
5 · What it builds	The deliverable decomposed into A, P and C, with the strict test for C.
6 · The verdict	How long before a competitor could replicate it; fund, fund with conditions, or do not fund — and what would build C.

The group presents step 1 and step 6, with the reasoning that connects them.

What to expect

Two results are common enough to be worth naming in advance, and neither is a failure of the project you chose.

Only one edge moves

Most project portfolios are built almost entirely of cost-edge projects. That is not wrong, but it is a choice, and usually an unexamined one. The question it raises: what would it take for this same project to also move willingness to pay — and would that cost much more?

C is empty

Projects deliver assets and processes because those are what can be specified, scheduled and signed off. Capability resists all three. The question it raises: if A and P are purchasable by any competitor, what in this portfolio is actually building advantage?

Download the completed canvas as a PDF, bring it to the afternoon session, and keep it. Module 3 closes with a group Execution Optimization Challenge, and an appraised project of your own is the kind of material it draws on.

TEN

Where the morning ends

The morning made one claim and spent three hours earning it: **operational efficiency is a consequence of what the customer values and of how work flows — not of how busy the resources are — and the returns to everything an organisation buys are scaled by a capability it cannot buy.**

The argument, in five moves

1. **Value is defined from the customer backwards.** Order winners create preference, qualifiers only avoid elimination, and the value stick says there are exactly two ways to create value: raise what the customer will pay, or lower what it costs to serve them.
2. **Waste is the gap between those two edges**, and the value stream map makes it visible: a few hours of work inside weeks of elapsed time, almost all of it waiting.
3. **The waiting is produced by utilisation, not cured by it.** Keeping everyone busy lengthens the queue by the physics of variability; flow efficiency, not resource efficiency, is what the customer experiences.
4. **Variation is the other half of the method.** Lean removes the time the work is not being worked on; Six Sigma narrows the spread that makes that time unpredictable. Neither holds its gain without the other.
5. **All of it shows up in return on capital** — in the margin through less rework, and in the denominator through less inventory. The denominator is the half of the equation that operational business cases forget.

Four equations, one staircase

Equation	What it adds to the one before
Value created = WTP − SOC	Where value comes from, and the only two moves that create it.
VC = $(WTP_A - C_A) - (WTP_B - C_B)$	Creating value is not having an advantage. Advantage is the difference — and a copyable improvement returns it to zero.
ROIC = margin × capital turnover	How the improvement reaches the people who fund it, through both levers at once.
V = C × (A + P)	Why two organisations with the same assets and processes get different results — and where the residue of advantage can live.

Lean Six Sigma is not a fifth equation. It is what climbs the staircase: Lean moves both edges of the first (waste out lowers cost; speed and reliability raise what the customer will pay), Six Sigma holds the gain through the third (less variation means fewer defects in the numerator and less inventory in the denominator), and DMAIC, repeated until it no longer depends on the team that ran it, is the coefficient in the fourth.

What should survive six months

- **Count elapsed time, not effort.** The question is never "how long does this take to do" but "how long does it wait" — and the answer is usually ninety per cent of the lead time.
- **Starting less finishes more.** Projects in flight divided by projects finished a year is your real lead time, whatever the schedules say, and it is the one number on the slide nobody can argue with.
- **Buy the bracket, build the coefficient.** Every tool and every method is available to a competitor at the same price. What is not is the organisation's repeated ability to find and fix its own problems — which is what Fremont changed, and what GM could not copy.

> **ON MONDAY**
>
> **ONE PROCESS** Take the recurring process you brought, walk it with the people who do it, and write two numbers on one line: hours of work, days of elapsed time. The ratio is the conversation you need to have.
>
> **TWO COUNTS** Projects in flight; projects closed in the last twelve months. Divide. Put the result beside the lead time your organisation believes it has.
>
> **ONE QUESTION** At the next approval you sit in, ask which edge of value the project moves and what is left when the competitor does the same. If the answer is nothing, fund it to the threshold and not beyond.

What the afternoon tests

The morning's identity says assets and processes are additive and purchasable, and capability is a coefficient. The afternoon takes that claim to the two places where organisations spend the most trying to buy capability: digital platforms, and other organisations' practices. If the identity is right, neither purchase should move the coefficient — and the afternoon's question is what does.

ELEVEN

Going further — how capability accumulates

This section is optional. The session closes on the claim that C is the multiplier and that it cannot be bought. The obvious next question — then how is it built? — is a research topic in its own right, and what follows is the short version of what the evidence supports.

Repetition alone does not accumulate capability

The learning curve — unit cost falling with cumulative output — has been documented since Wright's 1936 study of aircraft manufacturing. But the later literature found it is not a uniform law. Argote and Epple (1990) showed that organisations vary widely in how fast they learn, and attributed the variation to organisational forgetting, personnel turnover, knowledge transfer from elsewhere, and scale.

That finding moves the question. If accumulated experience does not determine learning speed, and organisations with comparable experience learn at different rates, the explanation lies in **what organisations do with the experience** — the deliberate component, not the automatic one.

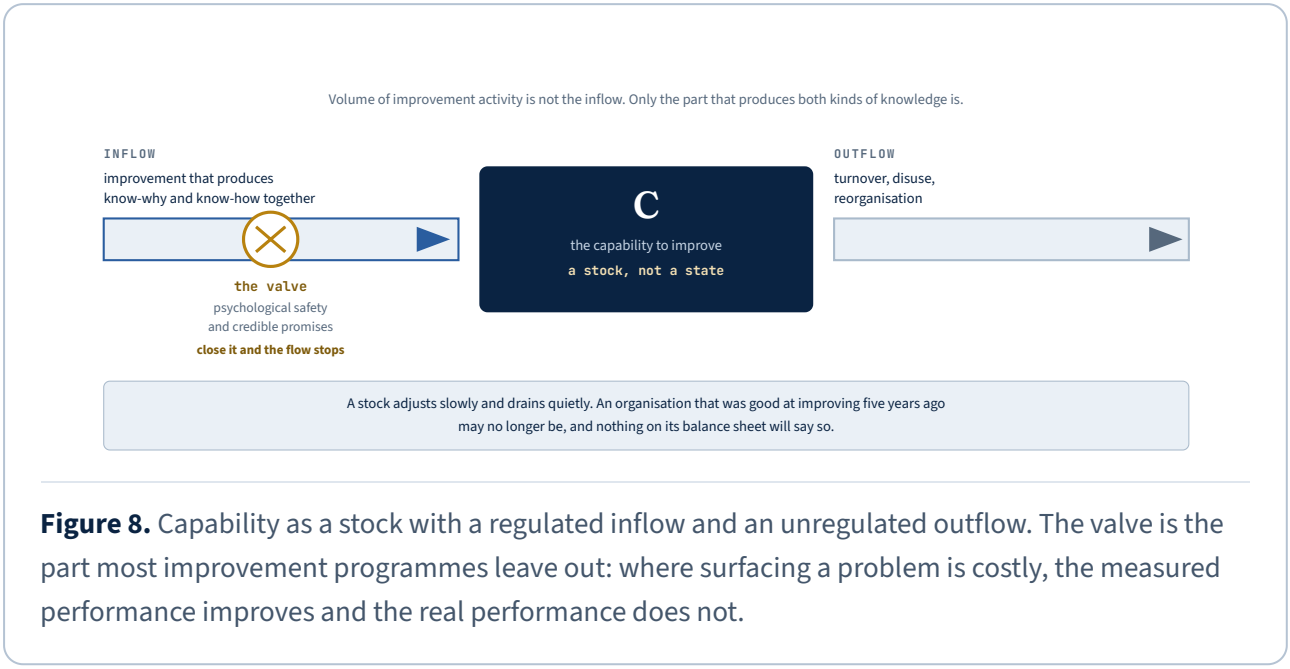


Figure 8. Capability as a stock with a regulated inflow and an unregulated outflow. The valve is the part most improvement programmes leave out: where surfacing a problem is costly, the measured performance improves and the real performance does not.

The finding worth remembering

Two studies of the same factory, a decade apart, give the sharpest decomposition available. Mukherjee and colleagues (1998), across 62 quality improvement projects, separated **conceptual learning** — understanding the cause-and-effect relationships, the *know-why* — from **operational learning** — validating the actions that produce the result, the *know-how*.

Lapré and colleagues (2000) then linked the learning activity of those projects to the factory's actual waste rate. The result is uncomfortable and precise:

> **LAPRÉ ET AL. (2000)**
>
> **Only about a quarter of the improvement projects accelerated the reduction of waste** — and they were the ones that produced know-why and know-how *simultaneously*. The remainder either slowed the reduction down or had no measurable effect at all.



Figure 9. The classification behind the finding, after Lapré, Mukherjee and Van Wassenhove (2000). Projects were sorted by whether they produced an understanding of the mechanism and whether that understanding was validated in the work. Only the cell that produced both accelerated the reduction of waste.

A project that produces know-how without know-why leaves the organisation able to repeat an action it does not understand. A project that produces know-why without know-how leaves it with a diagnosis nobody implemented. Neither accumulates anything. This is the empirical core of the DMAIC argument in Part Seven, and it explains why the Analyze phase is not a formality.

Pisano (1994) adds a boundary condition from 23 pharmaceutical process-development projects: where the underlying science is deep and well understood, learning *before* execution — laboratory experimentation — speeds development. Where it is not, as in biotechnology, it does not. The effective form of learning depends on how much is already known about the process.

WHERE THIS GOES

If improvement projects are how capability accumulates, two questions follow immediately: under what conditions do they work, and how is the accumulated stock lost again? Those are the closing questions of the *afternoon* session, after benchmarking has shown why the stock cannot be acquired from anyone else.

TWELVE

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